Global optimization of Savonius-type vertical axis wind turbine with multiple circular-arc blades using validated 3D CFD model

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ABSTRACT

Vertical-axis wind turbines, such as Savonius-type wind turbines (SWT), are an attractive choice for small-scale electricity generation. Some of the advantages of the SWTs are low speed, low noise, and self-starting ability, but the main disadvantage is the low efficiency. In this paper, the efficiency is improved by optimization of a multi-blade SWT (2, 4 and 6), with blades based on circular arc segments. Additional optimization variables were the tip-speed ratio and the aspect ratio. The main novelty of this paper is the global optimization process using genetic algorithm, which analyzed many different designs ($\sim 10^5$) using a validated 3D CFD model with 4eq SST turbulence model. As large number of designs was investigated, the result may be considered as the global optimal solution. In the 4 and 6-blade cases, the global optimal solution was similar to the previously proposed Scooplet-based design. The design was improved to power coefficient $c_P=0.32$ with aspect ratio 1.5, and additionally to $c_P=0.34$ with increased aspect ratio. For the 2-blade case, the obtained design was significantly different than the classical SWT and achieved 12% relative improvement.

Keywords: Shape optimization; Vertical axis wind turbine; Savonius wind turbine; Global optimization

1 INTRODUCTION

Savonius wind turbine rotor is simple in design and can be easily fabricated. The SWT operation is independent of the wind direction what makes the yaw system needless. SWT has the best starting torque characteristic of all wind turbines. The low value of the optimal SWT tip speed ratio TSR (0.7- 1.1) yields a significantly lower turbine rotational speed in comparison to horizontal axis wind turbine (HAWT) of the same size (with optimal TSR of 6-7). Consequently, SWTs are low-noise, cost–effective and require low installation space. This makes them competitive for microgeneration and for locations in urban environment or residential area.

Besides many advantages, the major drawback of SWT is its low peak power coefficient c_P of 0.2 (for classical Savonius rotor) considering that HAWTs achieve c_P =0.4-0.5 and VAWTs of Darrius type c_P =0.35-0.4. SWT has high (aerodynamic) solidity and pulsatile output with high amplitudes of torque-power per revolution. Through many modifications of the rotor geometry the value of the peak power coefficient was raised from 0.2 of the initial classical Savonius design to the value close to 0.3 while preserving the simplicity of design and high starting ability [1], [2]. The modifications of SWT for better power performance can be classified in the following categories: blade shape (S shape, Bach shape, Benesh shape, semi-elliptical, spline..), arrangement od blades (overlapping, spacing, tilting, spiraling, multi staging,…), multi-blades design. By adding separate stationary structures ("augmenters") around the SWT rotor (deflectors, shields, curtains, guide vanes….) the power coefficient of SWT is significantly increased, however at expense of the directional insensitivity and simple design [3].

The rise of interest in the Savonius design started in the 1970 and was accompanied with the series of experimental papers that resolved the effect of the rotor aspect ratio AR, rotor Reynolds number, number of blades and overlapping of blades. The experimental optimization of SWT performance was based on physical considerations of the flow in SWT [4]–[6]. Until 2000, only few papers analyzed theoretically the SWT flow by using the discrete vortex method DVM. First papers (or among the first ones) that used the numerical (CFD) analysis of the SWT two dimensional (2D) were published little after 2000 [7], [8]. 3D CFD simulation of the flow was performed for the SWTs with twisted blades in 2009 [9] and gradually evolved into a validation tool for the 2D optimization results.

In most of the CFD papers the SWT flow is modelled as a 2D flow which resembles the mid-plane flow between the rotor endplates. Basically, this corresponds to the flow in SWT with rotor aspect ratio $AR \nearrow \infty$. 2D prediction is a kind of upper bound for the SWT power performance prediction. 2D approach is advantageous due to the reduction of the computational time and computer resources required for the large number of CFD simulation in the optimization of SWT rotor geometry. Typically, 2D CFD simulations (most of which use RANS 2eq turbulence models, particularly the Shear Stress Transport [10]) over-predict c_P and the peak of c_P-TSR curve is slightly displaced to higher TSR in comparison with the experimental results. However, the same turbulence model under-predicts the c_P-TSR performance when used for 3D CFD modeling of the SWT flow [2], [11]. The transitional or 4eq SST turbulence model reduces the 2D CFD over-prediction of c_P for SWT flow at "low" rotor Reynolds number Re= $v_\infty D/\nu=10^5$ [12]. The use of

SST DES model, a hybrid of SST and Detached Eddy Simulation turbulence model, provides better agreement with the experimental cP values than the SST model [13] in 3D simulation.

The first modification of the original Savonius blade were based on modifications to the overlap ratio s/d (see Figure 1). In [4], [14], it was experimentally determined that optimal values are ratio near s/d=0.1. This was subsequently confirmed using 2D CFD in [15]. The effect of the end plate diameter was analyzed in [16] and it was found that using d_{ep}=1.1d (see Figure 1), improves the power coefficient. This value was used in most subsequent studies. In many studies, the effect of the rotor aspect ratio (AR=H/d) was considered since it has a significant effect on the rotor performance. In the experimental study [4], it was determined that the power coefficient increases by 10% from AR=1 to AR=1.5. Further rotor improvements were shown to be possible by changes to the blade shape. For example, Bach and Benesh SWT blades [17] are blade modifications that add a straight-line segment to the classical circular-arc blade. In one of the variations [1], this shape was shown to improve the power coefficient from 0.22 to 0.275. Later, many different modifications emerged, in which blades are based on splines [18] or elliptical curves [12]. These designs were analyzed numerically, and relative improvement was up to 12% compared to the classical SWT. Further blade variations are achieved by varying the number of blades, for example, multiple quarter blades were analyzed based on 2D CFD model in [19], [20]. In these studies, maximum power coefficient c_P=0.25 was obtained. A further improvement was achieved in the extensive optimization study where scooplet-based design (addition of smaller blade pair to a pair of larger blades), achieved c_P=0.36 using 2D CFD [2]. Also in [2], it was shown that 3D CFD predicts significantly reduced power coefficient c_P=0.31. Still, this was shown to be a 40% improvement over the classical SWT using 3D CFD.

The current paper investigates the optimum configuration of Savonius-type VAWT composed of multiple blades (2, 4 and 6). The novelty of the paper is that the conducted global optimization process investigated many designs (~10^5) using a validated 3D CFD model. As the large number of designs was investigated, the results regarding multiple circular-arc design generated based on CFD can be considered as global optimal solution.

2 VERTICAL AXIS WIND TURBINE MODELING

2.1 Vertical axis wind turbine performance

This section describes important definitions used in the subsequent optimization and analysis. The main operational parameter of SWT is the tip speed ratio (TSR or λ), defined by:

$$TSR = \lambda = R\omega / v_\infty \tag{1}$$

where ω is the angular velocity. For the Savonius type VAWTs, the optimal TSR is usually near λ=1. In this paper, TSR value was selected as optimization variable within range from 0.5 to 1.5. The most important value for evaluation of the SWT performance is the power coefficient:

$$c_P = P / (\rho v_\infty^3 A / 2) \tag{2}$$

where the generated power P is obtained from numerical analysis, for a selected fluid density ρ, and wind velocity v. The rotor swept area A is defined as A=2RH, where R and H are the rotor radius and height respectively (see Figure 1).

The power P is calculated by averaging the generated torque over the rotor revolution. In CFD analysis, the average torque is calculated for each time-step by integration of forces generated by the fluid on the rotor surface A_b. The forces can be divided to pressure p_w and shear stress τ_w forces. The overall effect of the forces produces net-torque which changes in time during the rotor revolution. The instantaneous torque depends on the rotor angle ϕ. For each blade, the torque $T_{b,i}$ can be calculated as:

$$T_{b,i}(\phi) = \int r dF_T = \int_{A_b} \left\{ \left[\vec{r} \times (p_w dA_b)\vec{n} \right]_z + \left[\vec{r} \times (\overrightarrow{\tau_w dA_b}) \right]_z \right\} \tag{3}$$

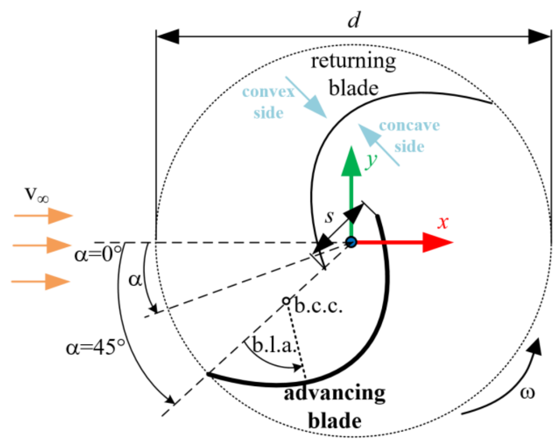
where dA_b is surface element with surface normal $\vec{n}$, $\vec{r}$ is the position vector and z is the axis of rotation. Full torque average T can be calculated by averaging the torque of each blade and summing over all N blades:

$$T = \sum_{i=1}^{N} \int_0^{2\pi} T_{b,i}(\varphi) d\varphi / 2\pi \tag{4}$$

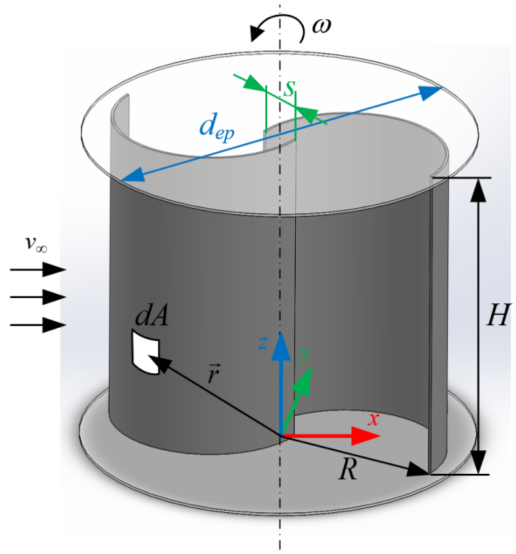
Finally, the VAWT power (averaged) is calculated by expression:

$$P = T\,\omega \tag{5}$$

This value is obtained based on numerical simulation and inserted into (2) to obtain the power coefficient. Closely related value is the torque coefficient averaged per rotor revolution, calculated as c_T=c_P/λ.



a)



b)

Figure 1. Main parameters of Savonius wind turbine: a) 2D section and b) 3D view.

2.2 CFD model

SWT modeling is almost always performed using computational fluid dynamics (CFD) [21]–[23]. These models are based on transient Reynolds averaged Navier-Stokes (RANS) equations. The equations are obtained by averaging the standard continuity and momentum equations. Some simplifications can be used when the fluid is practically incompressible (as is in SWT modeling case). In Cartesian coordinate system, the equations can be compactly written using index notation:

$$\frac{\partial \overline{u_i}}{\partial x_i} = 0 \tag{6}$$

$$\rho\frac{\partial\overline{u_i}}{\partial t}+\rho\frac{\partial}{\partial x_j}\left(\overline{u_i}\,\overline{u_j}+\overline{u_i'u_j'}\right)=-\frac{\partial\overline{p}}{\partial x_i}+\frac{\partial\overline{\tau_{ij}}}{\partial x_j}+\rho f_i \tag{7}$$

where x_i are the space coordinates; the time is designated by t; the density, velocity and pressure (ρ , $\overline{u_i}$ and p) in the equations are the Reynlods-averaged flow values; in the case of rotating reference frame, specific body forces f_i have to be included (the Coriolis and centrifugal forces); $\overline{\tau_{ij}}$ are the time-averaged (mean) viscous stress tensor components:

$$\overline{\tau_{ij}}=\mu\left(\frac{\partial\overline{u_i}}{\partial x_j}+\frac{\partial\overline{u_j}}{\partial x_i}\right) \tag{8}$$

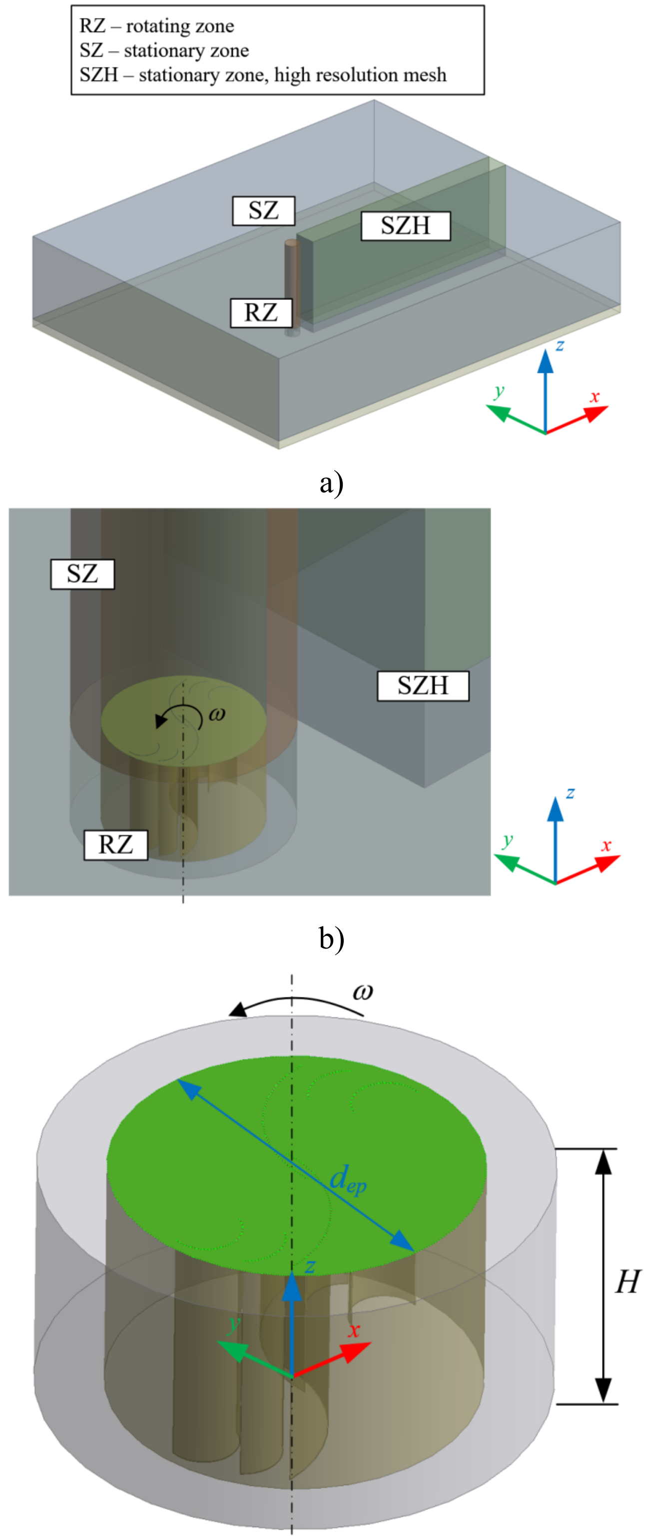
where the fluid viscosity is designated as μ . The Reynolds stresses - $\rho\overline{u_i'u_j'}$ in equation (7) are modeled according to the Boussinesq eddy-viscosity assumption, R_{ij} :

$$R_{ij}=-\rho\overline{u_i'u_j'}=\mu_t\left(\frac{\partial\overline{u_i}}{\partial x_j}+\frac{\partial\overline{u_j}}{\partial x_i}-\frac{2}{3}\frac{\partial\overline{u_k}}{\partial x_k}\delta_{ij}\right)-\frac{2}{3}\rho\delta_{ij}k \tag{9}$$

The turbulent viscosity μ_t and turbulence kinetic energy k are obtained from the selected turbulence model. In most SWT numerical studies, SST two-equation model is used, since it demonstrated decent prediction accuracy in various studies (for example [24], [25]). An improvement of the SST model can be achieved by using 4-equation transitional SST turbulence model, which was the model used in this paper.

The computational domain shape and size was based on previous research [2]. Figure 2 shows the overall domain and zones. One half of the rotor is included in the computational domain, with a symmetry condition applied on the rotor mid-plane. The rotational zone (RZ) includes the rotor and it has a slightly larger diameter than the rotor (1.8*d, d*=0.68m). The rest of the computational domain is divided into several stationary zones. The domain dimensions are somewhat increased from the dimensions in previous research [2]. The overall dimensions of the computational domain are: height in *z*-direction is 5*d*, *x*-direction length is 40*d* and the *y*-direction width is 30*d*. A part of stationary domain related to SWT wake is separated from the main for the purpose of further mesh refinement in that region. This part of stationary domain with higher mesh resolution has 3*d* width, it starts 2*d* downwind from the rotor axis and extends to the domain outlet. The rotor axis is positioned 15*d* from the inlet section and 25*d* from the outlet section.

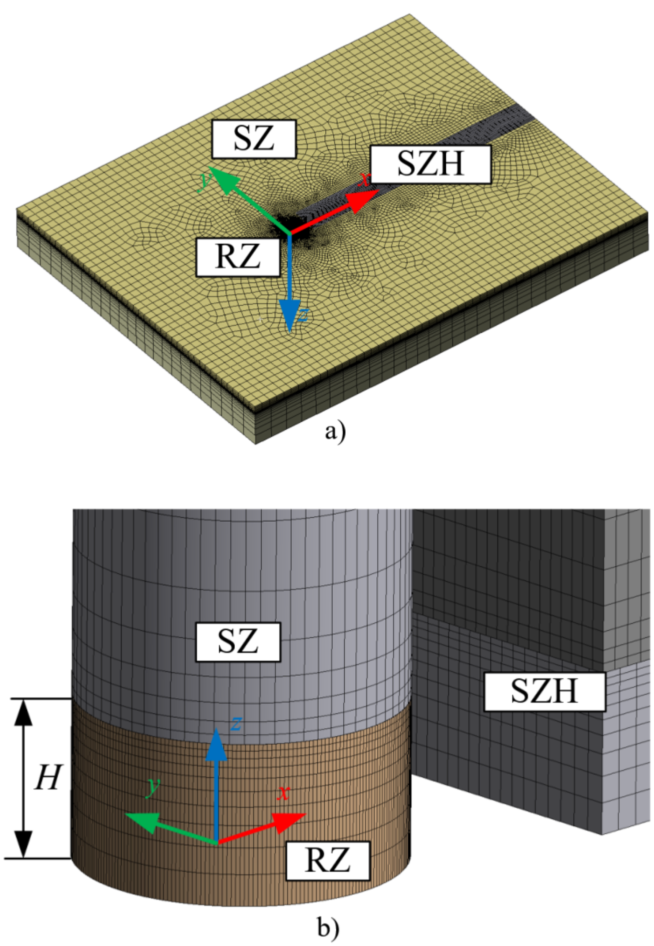
In the initial case, the wind velocity and rotational speed were constant, set to v_∞=12 m/s and ω=31.4 rad/s, which corresponds to TSR=0.9. However, during the optimization the TSR was used as an optimization variable. Between the rotational zone and the stationary zone, sliding mesh interface is implemented. At the outlet boundary, pressure was set to zero. On the lower (–*z* direction) plane a symmetry condition was applied, since this corresponds to the rotor mid-plane. Furthermore, the same symmetry condition was applied to the remaining sides (+*z* direction an ±*y* directions).



c)

Figure 2. Computational zones: a) overall computational domain b) stationary zones near rotor and c) rotor zone

The mesh was fully structured mesh only in the SZH zone. In the remaining zones, the mesh was unstructured regarding the 2D shape (in *x*-*y* plane), and the sweep operator was applied in the *z*-direction. The 2D shape was unstructured quadrilateral mesh. In the rotating domain, a thin structured prismatic layer was used near the blade walls. Blade edges include further refinement, see Figure 3c. The prismatic layer was composed of 20 mesh layers in the direction normal to the wall surface. First layer thickness was 0.01mm so that condition y+≈1 is satisfied. The blade-edge refinement (LTE) corresponds to 0.5 mm mesh element size. On the rest of the blade (BL), mesh element size was 5mm, and 13mm was the element size in the rest of the stator domain. In the stator zone (SZ), 600mm elements were used, with a smooth transition to smaller elements in the adjacent zones (growth rate 1.1). In the SZH zone, element size was 125mm, with a gradual increase to 600 mm towards the domain outlet. By height, total of 20 elements were used of which 10 in both lower and upper parts of the domain. The mesh was refined by height so that the smallest element size is near the rotor edge-plate. These were mesh settings related to the "high-resolution" mesh, and the scaling factors in the following grid independency study is related to all of the mentioned element sizing parameters.



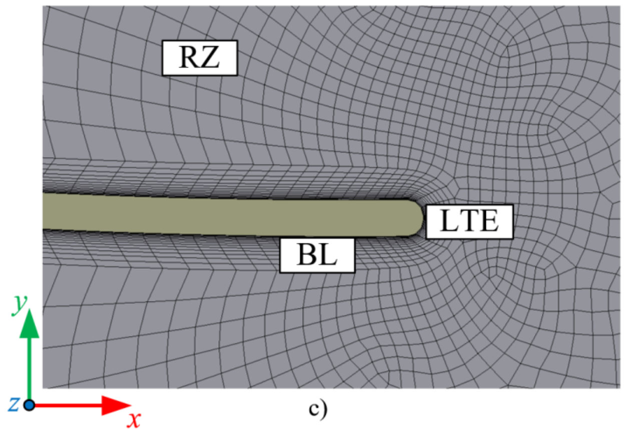


Figure 3. Mesh elements at different computational: a) overall domain, b) near-rotor zones and c) prismatic layers and blade tip refinement.

2.3 Grid independency study

For grid independency study, two SWT designs were used. First one is the classical semicircular SWT at TSR=0.9, Table 1a. The second design is an intermediate-optimized 6-blade design, Table 1b. Starting from previously described mesh, the element size was uniformly scaled by ~1.3 and the time-step was scaled by factor 2. For the classical SWT design, the solution remains almost constant for wide range of mesh and time-step resolutions as shown in Table 1a. The lowest resolution mesh contained 175k elements and the highest contained 1.6M elements. Table also includes results from previous study [2] in which the mesh contained 20M elements. Since this was an uneventful test, an additional test was performed with a more complex 6-blade geometry (final optimized design will be presented in the result section). The resulting power coefficient is presented in Table 1. The difference between the highest and lowest resolution test is 3%, and the difference between the last two is only 1%. Since the highest resolution mesh was used for evaluating all the results, the mesh related error is expected to be 1%.

Table 1. Mesh test results

Mesh elements	Time step (°)	c_P (-)
175119	4	0.2201
314767	2	0.2189
437140	1	0.2173
509140	0.5	0.2168
886938	1	0.2184
1586786	1	0.2183
20M[2]	1	0.2187

a)

Mesh elements	Time step (°)	c_P (-)
118358	4	0.3233

208242	2	0.3199
358822	1	0.3179
511116	1	0.3145
511116	0.5	0.3143

b)

Another important value which needs to be converged is the moving average of the power coefficient between subsequent rotor revolutions. For the highest resolution mesh, Figure 4 shows the instantaneous torque coefficient c_T for two final rotations, together with the moving average of the instantaneous $c_T(\alpha)$. As it can be seen, the variations of the moving average are negligible. Also, the interval of averaging can be arbitrary since there are no differences between half-revolution average and the 2-revolution average. In this paper, the 2-revolution average is used for evaluation of the power coefficient.

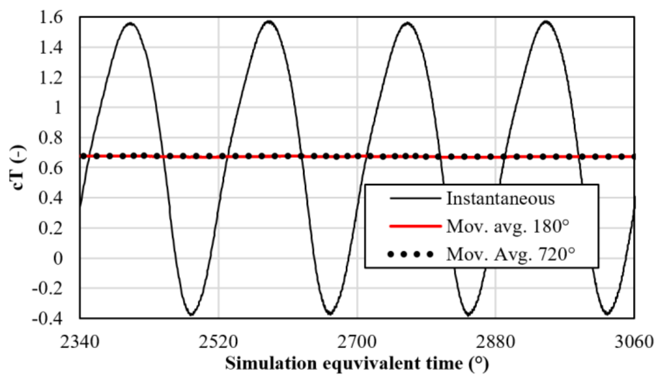


Figure 4. Instantaneous torque coefficient (c_T) and moving average of instantaneous $c_T(\alpha)$ for half-rotation (180°) and for two rotations (720°).

2.4 CFD model validation

For validation, two experimental data sets were used. The first is related to integral values (power coefficient) and the second is related to local values (pressure coefficient). A reliable and commonly used power coefficient curves for different SWT designs can be found in the research paper [4]. Among many results from the paper, the design with the highest power coefficient is used here for validation. This SWT design uses semi-circular blades and the overlap ratio is equal to s/d=0.1.

Several turbulence models were investigated in the current paper, and similar results were achieved in all cases. The CFD model was implemented in the ANSYS FLUENT 2019R3 software [26]. It can be seen in Figure 5 that current 3D CFD with SST model provides almost the same results as the previous 3D CFD model in [2]. The difference between these two is in mesh type and the number of elements. In the current paper, the mesh is based on hexahedrons and ~0.5 million elements were used, while the model in [2] used polyhedral mesh with ~30 million elements. This result again confirms that the selected mesh is adequate. Comparison of SST and 4 equation transitional SST model shows only a minor difference. Compared to experimental data [4], all models predict slightly lower power coefficient values, among which the 4-equation SST model achieves marginally better prediction than the 2-equation SST model. All of the remaining results in this paper were based on the 4-equation SST model.

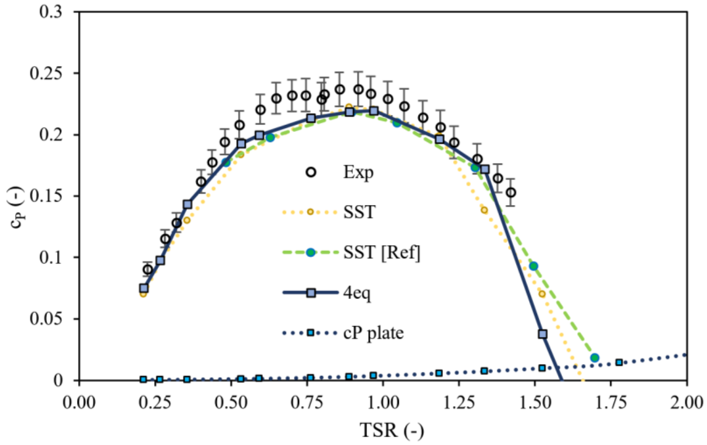


Figure 5. Classical Savonius wind turbine (*s*/*d*=0.1) performance, comparison of experimental [4] and 3D CFD results for different turbulence models in current paper (SST and 4-eq) and from previous research (Ref. [2]).

For validation of local values, a CFD model was created using data from [14]. The design was similar to the classical Savonius, but with blade overlap s/d=0.15, AR=1, and the rotor diameter is 300mm. The design is characterized by significant blade thickness amounting 4mm. The experiment was conducted in wind tunnel at v_∞=6m/s, and TSR=0.9, and no correction was applied to measured data due to the tunnel blockage effect. The local pressure values was measured at several points on the turbine mid-section. To simplify the comparison and reduce the errors, only the pressure difference was compared. The pressure difference is calculated as a difference between the same points on the blade convex and concave sides. Instead of pressure, pressure coefficient is compared in Figure 6. The advancing blade is the one with 0-180° local angle, and the returning blade is the one with 180-360° local blade angle. See Figure 1 the for definition of blade local angle (b.l.a.) with respect to the blade center of curvature (b.c.c.). The differences between the numerical and the experimental results are mostly related to the area of lowest pressure on the convex side of the advancing blade.

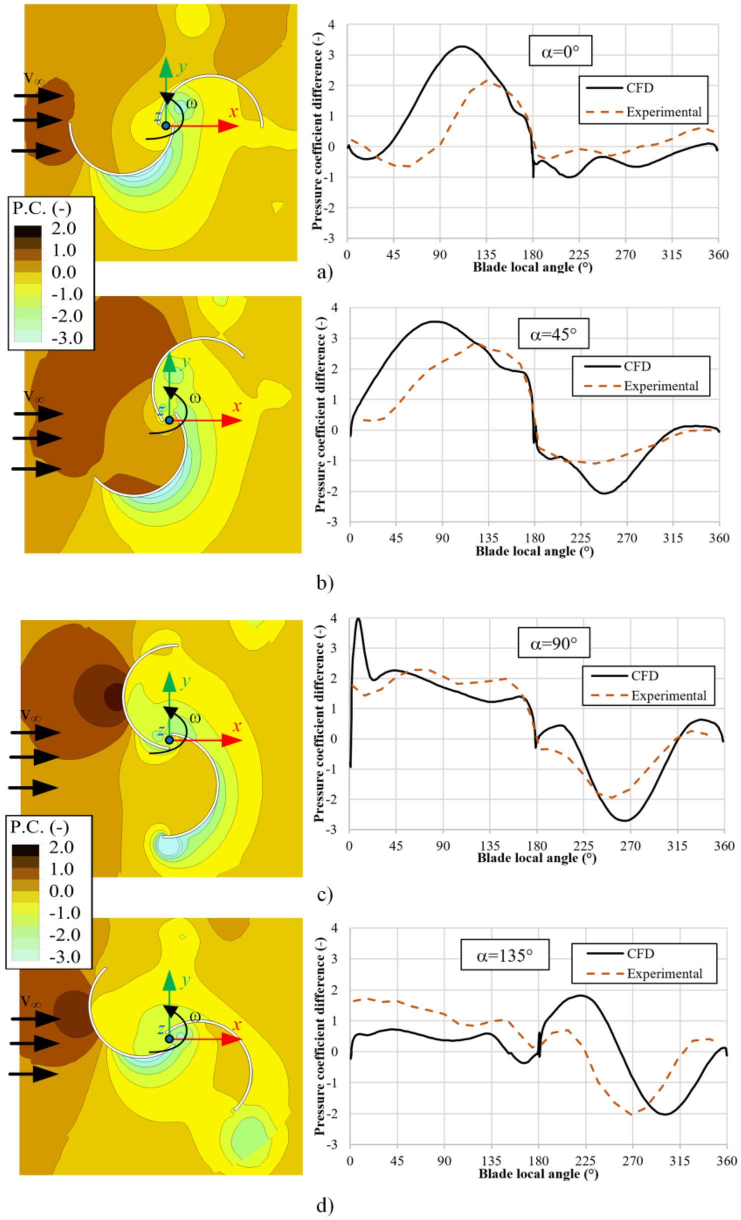


Figure 6. Comparison of CFD with local pressure difference (between convex and concave blade side) using geometry and experimental data from [14] for rotor angles: a) α=0°, b) α=45°, c) α=90° and d) α=135°.

3 SHAPE PARAMETERIZATION AND OPTIMIZATION

3.1 *Shape parameterization*

Advanced shape parameterizations are usually based on NURBS or B-spline based shapes which allow abundant shape flexibility. This would allow the optimizer to generate novel complex designs as discussed in [27]–[29]. The problem with these shape parameterizations is related to the large number of design variables. In transient CFD-based optimization studies, the number of shape variables should be reduced as much as possible. The shape parameterization can be considered efficient if a wide variety of shapes can be represented by using only a small number of design variables. In past Savonius optimization studies, different parameterization methods were applied to reduce the number of design variables. In [30], a polar coordinate system is used to define the blade shape. In [25], several models based on B-spline curves are presented for reducing the degrees of freedom for individual spline control points. The blade design can also be based on elliptical sections [12] or in a general case by a single spline curve [18]. A simplification (reduced number of shape variables) can be achieved by Bach-like blades [17], which are described by a straight segment and a circular part – but only a few selected variables were used in previous research. An arbitrarily positioned circular arcs (up to 2 different arcs) were used in [2]. In the current paper, also circular arcs were used, but up to 3 different arcs were selected. The additional optimization variable was the TSR, and the influence of the aspect ratio was afterward considered.

Up to 3 blades (labeled c1, c2, and c3 in Figure 7) are independent in the optimization as shown in Figure 7. The remaining "dependent" blades are always the same shape as the three independent blades, only rotated by 180°. The individual blade is defined with up to five design parameters: x-y coordinates of both blade edge (4 parameters) and blade curvature radius which is parameterized as the blade "thickness" by parameter m_i (see Figure 7). One point was kept fixed in all optimization cases since the rotation of the blade as a rigid body is not important. The point was fixed at the outer rotor radius which was constrained in the optimization. A constrain is added to prevent designs with intersecting blades, however, they can still be arbitrarily close to each other, as for example shown in Figure 7.

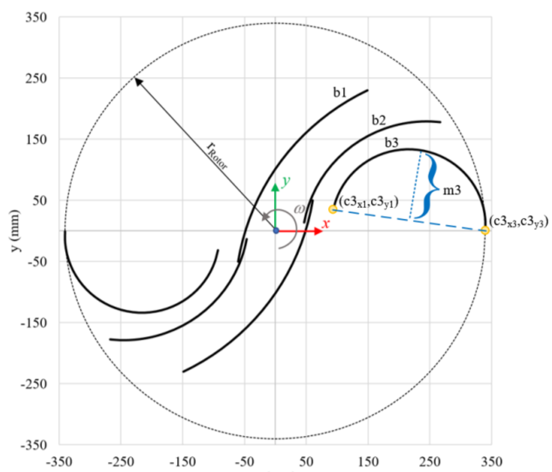


Figure 7. Shape parameterization composed of circular arcs.

In all of the performed optimization studies, the final shape is not known so it is important to assure mesh quality is good in all cases. Many random test cases were analyzed to confirm that this is indeed the case. To show the various designs which are included in the design space and to illustrate the wide variety of designs that are to be analyzed to find the global optimum, some of these designs are shown in Appendix A.

3.2 Optimization procedure

The average power coefficient ($c_{P,ave}$) was the optimization objective in the single objective optimization problem. During optimization, the mesh with equivalent time-step 1° (see Table 1) was used. Shape variables were the geometry-related variables as described in previous section (Section 3.1) and the rotational speed (TSR). In the simplest 2-blade case (1 independent blade + 1 rotated blade), there was a total of 4 optimization variables. For 4-blade case (2 independent blade + 2 rotated blade) there were 9 optimization variables and finally 14 optimization variables for the 6-blade case (3 independent blades + 3 rotated blades). The final variable was the aspect ratio, which was analyzed in a separate section. In previous optimization studies, the TSR was usually fixed to a constant value, most commonly the selected design was TSR=0.9. In this study, TSR was an optimization variable within the range from TSR=0.5 to TSR=1.5.

The optimization procedure was implemented using ModeFrontier [31]. The NSGA-II variant of the genetic algorithm was used. Since genetic algorithms can be parallelized, the computational procedure was performed on 8 PCs with (slightly different configuration), which enabled parallel execution of 18 CFD simulations. The starting point of the optimization procedure is selecting the initial population, for which the Sobol sequence [32] was used. The size of the number of SWT designs in the first generation was 50, 500 and 2000 for the optimization cases with 2, 4, and 6 blades respectively. The overall results of all three optimization runs are shown collectively in Figure 8. Notice that the number of design evaluations till convergence for cases with 2, 4, and 6 blades is progressively larger by an order of magnitude.

The optimization was conducted in two stages. First, the optimization was run with the default settings. The results were monitored continuously. When no drastically new shapes appear, the optimization run was continued using reduced population size and reduced mutation rate by half. To confirm the results, the optimization process was repeated multiple times for all cases, but with different initial populations. The obtained optimum was the same in all cases, except in some cases when the initial population size was reduced.

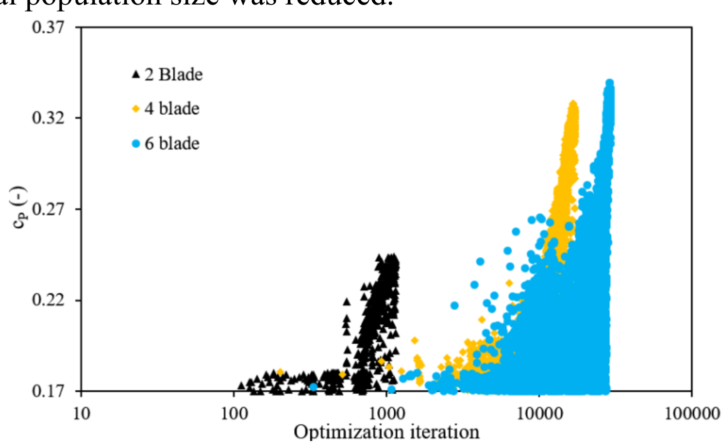


Figure 8. Convergence of power coefficient for global optimization cases with 2,4 and 6 blades.

4 RESULTS AND DISCUSSION

4.1 *Overall results*

All the result in this section were obtained using the same 3D CFD model, with 500k elements, and 0.5° equivalent time step, which was shown to be converged in the mesh independency test section (~1% error). Aspect ratio was AR=1.5. The power coefficient characteristics of the 2-blade design can be noted first. The power coefficient is improved by 12% relative to the classical SWT with overlap ratio s/d=0.1. Most importantly, the global optimization process has adjusted the operating rotation speed to TSR=1.2. This result is a significant change from the most commonly used range between 0.9 and 1.0. The remaining two designs have an alike power coefficient characteristic. The 6-blade case obtains the highest power coefficient. However, this is only slightly (almost negligibly) compared to the 4-blade case. This result suggests that a further increase in blade number would not yield an additional improvement. The improvement over classical s/d=0.1 design was 48%.

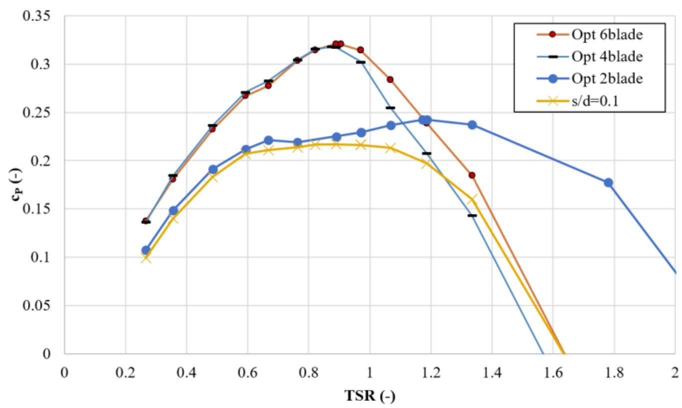


Figure 9. 3D CFD comparison of c_P-TSR of globally optimized blade (2, 4, and 6 blade) with semicircular 2-bladed design with s/d=0.1.

4.2 *Optimum for a 2-blade design*

This case contains a total of 4 optimization variables: TSR, *x* and *y* locations of the inner blade point ($c3_{x1}$ and $c3_{y1}$) and the blade camber from the chord line (*m3*).

Typically, in previous studies, the TSR is fixed to a predetermined value TSR≤1. When using 2D CFD-based optimization with the fixed TSR, the resulting optimal shape is similar to the classical Savonius design [2]. In the current paper, it is interesting to note that the optimized shape is significantly different from the classical one. This is especially true regarding the overlap (s/d) which is practically zero here, instead of typical values near s/d=0.1.

The optimized geometry in the current paper obtains power coefficient 0.242. Meanwhile, the classical Savonius design achieves power coefficient 0.217 when evaluated using the 3D CFD model used in this paper. This design achieves 12% improvement in power coefficient while also potentially reducing the production cost since the blades are slightly flatter.

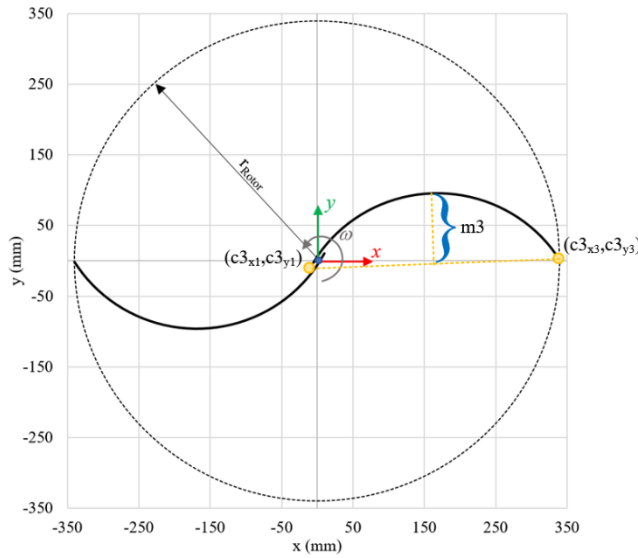


Figure 10. Optimized 2-blade design

The optimized TSR value for this case was TSR_{opt}=1.2. To analyze the design sensitivity, Figure 11 shows the parallel coordinates chart in which only the best designs from the optimization study are shown (the remaining designs are shown in pale-gray). The best design is shown by a thick line. The color gradient shows the sensitivity to deviations from the optimized design. It is seen that the TSR can be changed from 1.1 to 1.3 without much changes. The geometric parameters can be changes within ~10 mm while keeping the same performance.

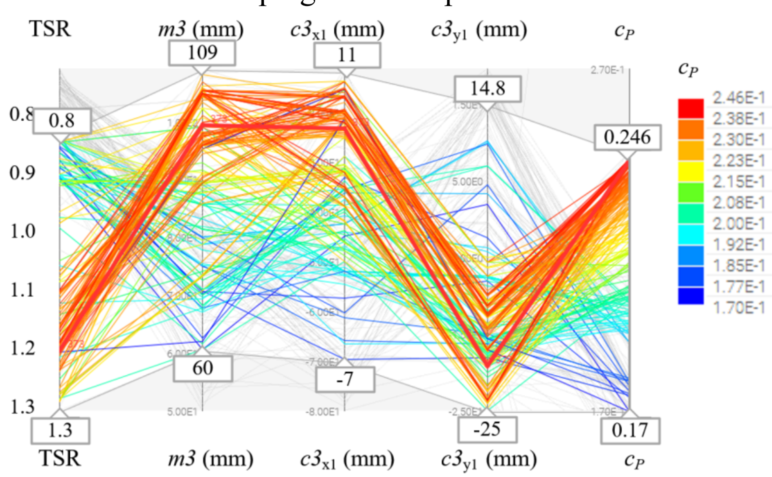
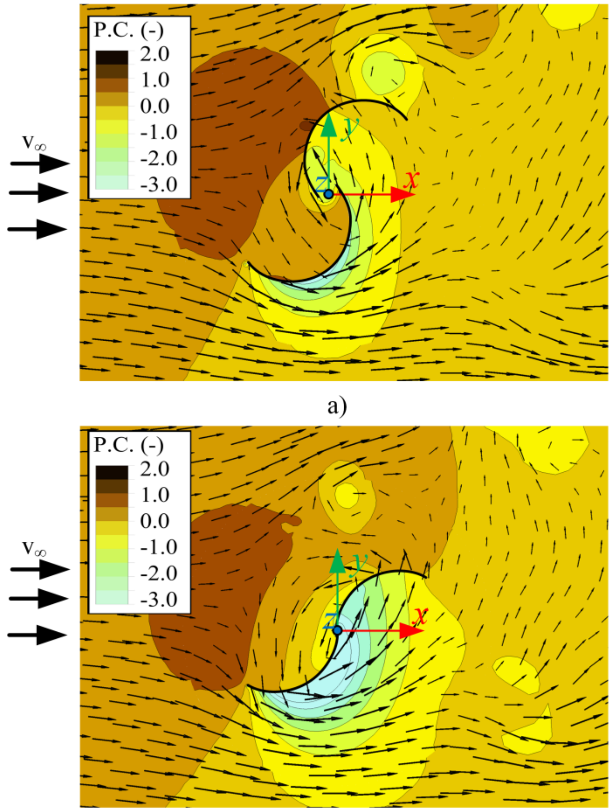


Figure 11. Parallel coordinates chart from 2-blade design optimization (near-optimal designs).

Figure 12 shows the flow field for the classical Savonius turbine and the optimized 2-blade design for the same α=45° angle. This is the angle which approximately corresponds to the maximum instantaneous torque coefficient, so it is appropriate to analyze the gains achieved by the optimized design. As it can be seen when comparing the two flow fields, the main difference is in the size of the negative pressure on the convex side of the advancing blade. This negative pressure acts positively on the net torque. Also, the returning blade of the optimized design has a lower positive pressure on the convex side. This also provides a net positive influence on the rotor torque.



a)

b)

Figure 12. Flow field for α=30° for a) classical Savonius with overlap s/d=0.1 and b) optimized 2-blade design.

4.3 Optimum for 4-blade design

Next figure shows the global optimal solution. This solution is achieved when the initial population is sufficiently large. This design achieves c_P=0.318. Compared to previously optimized design which used 2D CFD during the optimization [2], this design has flatter main blades i.e. increased curvature radius. It can be noted that the spacing between the main blade and the smaller (scooplet) blade is almost the same in both designs. When these two designs were compared using the same 3D CFD model and the same aspect ratio, the new shape achieves a 3% relative improvement.

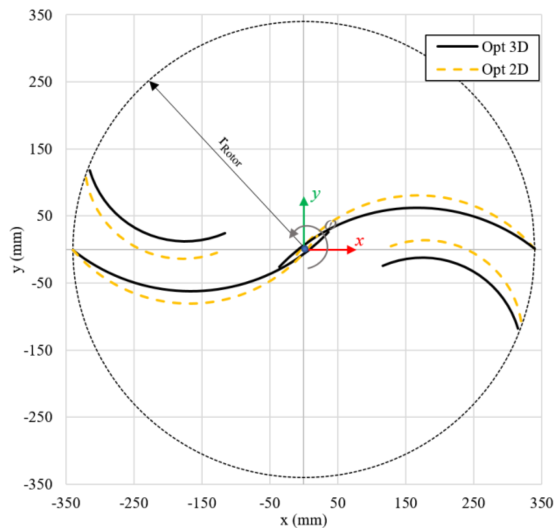


Figure 13. Global optimum for 4-blade design (3D CFD) and comparison with previously optimized design using 2D CFD [2].

To analyze the contribution of the 3D CFD compared to 2D CFD, Figure 14 shows the distribution of the power coefficient by rotor height. For the classical Savonius turbine and the 2-blade design the results are as expected. The highest contribution to the power coefficient are in the rotor mid-plane (symmetry plane), with a gradual decrease towards the end-plates (at *z*=0.5m). For the optimized 4-blade and 6-blade designs, the results are somewhat unexpected. The maximum contribution of the rotor to the power coefficient is not located in the blade center but about half-way between the mid-plane and the edge-plates (at *z*=0.33m). Since the 3D was validated regarding local pressure values (see section on CFD validation), these local-values could also be regarded as reliable. This fact justify the application of 3D CFD during the optimization, since 2D CFD cannot predict these effects.

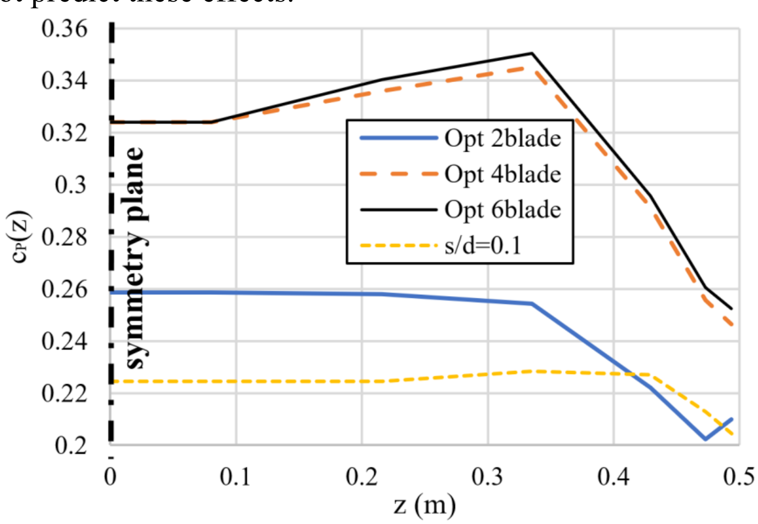


Figure 14. Blade contributions to power coefficient as function of height for different designs at corresponding optimal TSR values.

The shape of the previous $c_P(z)$ curves for 4-blade and 6-blade designs has an unexpected location of the maximum. The expected shape would contain maximum in the symmetry plane and a steep decline near the end-plates (z=0.5), as is the case of 2-blade design. In any case, this would be true for 2D CFD, which is equivalent to large AR design. To confirm the validity of previous results, 3D CFD analysis was performed using different AR. As the AR increases, the curve should become flatter and decrease steeper near the end-plates. Figure 15 confirms these expectations for the 6-blade optimized design.

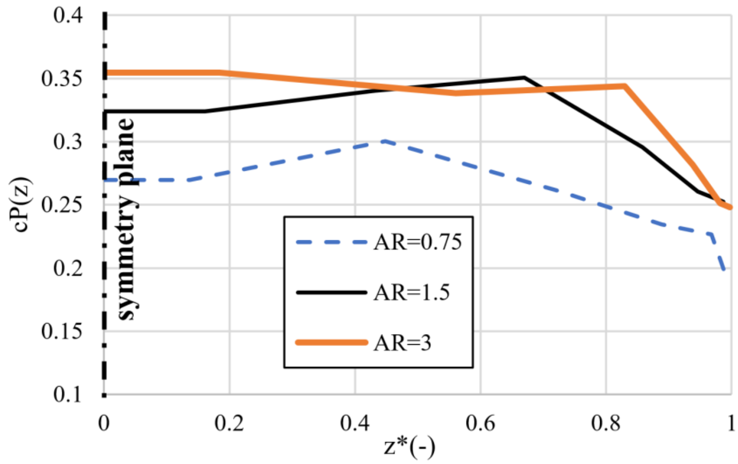


Figure 15. Optimized 6-blade design: Power coefficient as function of height for different aspect ratios, (z*=z/(H/2)).

4.4 *Optimum for 6-blade design*

Figure 16 shows the global optimal solution for the 6-blade design. This optimization was repeated several times, and the global optimal solution is achieved only when the initial population was sufficiently large. This design achieves c_P=0.321, which is a slight improvement over the previous design [2]. The basic design is similar to the previous one with addition of one smallest blade. If this trend would to be continued when the number of blades is increased, this would generate increasingly smaller blades, and the improvement between designs would become negligible. The most important conclusion regarding this design is that further increase in the number of blades is not likely to increase the turbine performance.

The 6-blade optimized design is also compared with commonly investigated multiple quarter blade design (MQB) [20], see Figure 16. Although the designs are topologically the same, the geometrical difference is extensive. The largest inner-blade of the MQB design is based on the classical SWT with overlap ratio s/d=0.1. The smaller quarter-blades in previous MQB optimization studies, are positioned arbitrarily and only few basic parameters are optimized, such as arc-angle and distance between blades. The maximum power coefficient of the MQB design in [20] was c_P=0.227 at TSR=0.73. The re-evaluation of the MQB design using the CFD model in the current paper resulted in maximum c_P=0.20 at TSR=0.76. Compare these values with the optimized 6-blade design which achieved maximum c_P=0.321 at TSR=0.9. This comparison clearly shows the significance of the proposed global optimization which allowed 60% power coefficient improvement over the multiple quarter blade design.

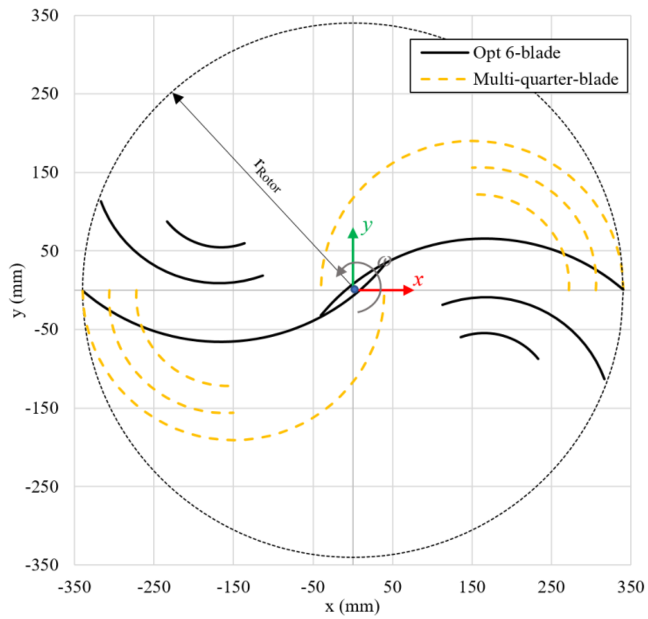


Figure 16. Global optimum for 6-blade design compared to multiple quarter blade design [20].

Local pressure coefficient values for the optimized 6-blade and the MQB designs are shown in Figure 17. Note that only one half of the rotor is simulated and the results are mirrored for clearer presentation. The rotor angles are selected to be the same as in [20], in which 3D CFD analysis of the MQB was also performed. The main difference between these two designs can be seen on the convex side of the advancing blade for α=0° and α=60°. Both the area and the intensity of the negative pressure are increased. But also, as it can be seen for α=60°, the area of maximum negative pressure is closer to the outer radius for the optimized 6-blade design. Some 3D effects can be observed in both design. Most notably, the pressure field changes near the end plates.

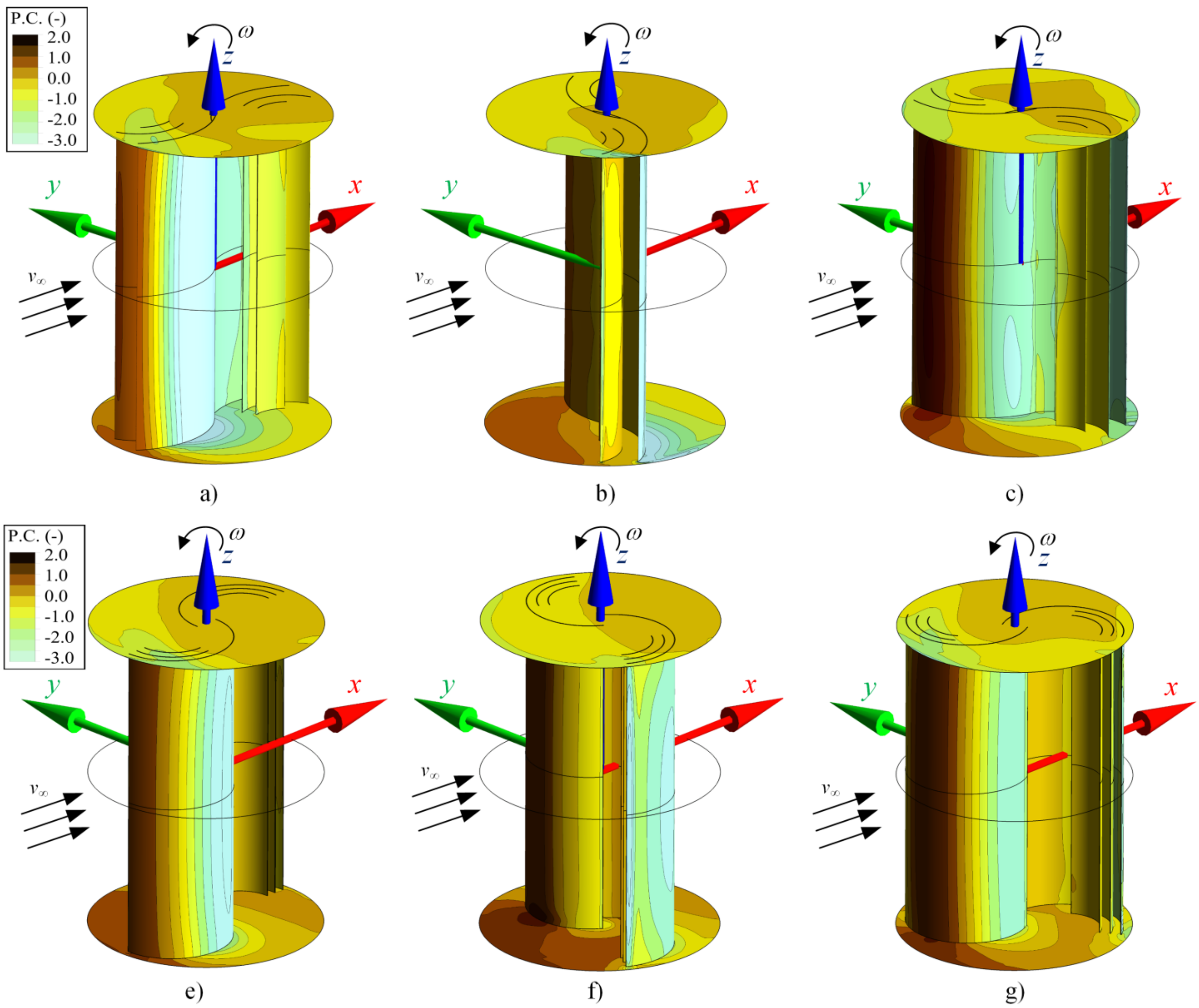


Figure 17. Contours if local pressure coefficient for optimized 6-blade design: a) α=0°, b) α=60°, c) α=120°, and multiple quarter blade design d) α=0°, e) α=60°, f) α=120°,

4.5 Optimal aspect ratio

The effects of aspect ratio (AR=H/D) on the power coefficient are shown in Figure 18. It can be seen that the selected AR=1.5 is not the optimal one, but it is near the maximum, which was obtained at around AR=3 in all cases. Further increase in aspect ratio shows only negligible c_P increase. For the 6-bladed optimized design, the power coefficient can be increased from 0.32 to almost 0.34. Similar conclusions can be stated for both 4-blade and 2-blade optimized designs.

In previous research [2], the scooplet-based was optimized using 2D CFD, which effectively corresponds to infinite AR. The power coefficient obtained from 2D CFD was 0.359, which was reduced to 0.304 when 3D CFD with AR=3 was used. Thereby the differences are explained due to application of 3D flow equations.

The 4-blade and 6-blade design are more sensitive to the aspect ratio, with 15% power coefficient increase from AR=1 to AR=2.5. Meanwhile, this amounts to only 5% increase for the 2-blade design. For AR<1, the power coefficient drops significantly for all designs.

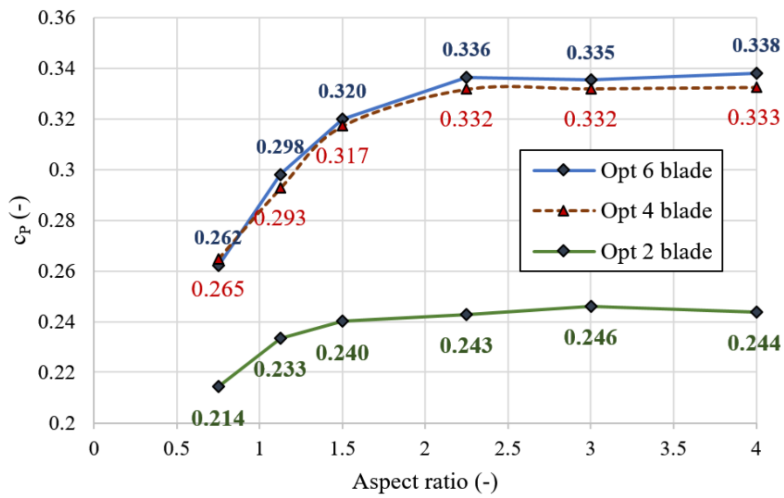


Figure 18. Effect of aspect ratio on optimized design power coefficient

5 CONCLUSIONS

This paper performed a global optimization of Savonius wind turbine (SWT) using a validated 3D CFD model. The 3D CFD model was validated regarding the power coefficient and also for the local pressure distribution. The genetic algorithm-based optimization was characterized as a global optimization since the large number of designs was investigated (~10^5). It was shown that large number of design evaluation was required in order to achieve blade shape convergence. Geometrically, the study was limited to using only circular-arc segments and was divided into three cases: 2,4 and 6-blade. However, even with this limitation it was shown that an extremely large number of different shapes exist. By repeating the optimization process several times, the global optimum was found in all three cases.

In addition to the shape optimization of blade, the paper has shown the importance of including the tip speed ratio (TSR) as an optimization variable. This was particularly true for the 2-blade case, in which a novel design was achieved which operates at TSR=1.2. This design showed 12% improved efficiency compared to the classical design, while also saving material costs.

In 4 and 6-blade cases, the global optimum was similar to the already known design, but the design was improved by 3%, and an additional 5% when considering the aspect ratio (AR), reaching a power coefficient of 0.34. Overall, this was shown to be an almost 50% improvement over the classical Savonius design. The aspect ratio was also shown to be important for these two optimization cases (more significantly than for the 2-blade case). The power coefficient improved by 15% from AR=1 to AR=2.5. The analysis of height-wise distribution of the power coefficient has shown that 3D effects are significant which demonstrated the importance of using 3D CFD model compared to 2D models.

Due to the promising results of the study, it can be expected that future optimization work using different geometrical shapes could yield further improvements. Instead of circular arcs, ellipse-based shapes or more general spline-based shapes could be used. The problems to be solved before attempting these studies will be the reduction of shape variables that are required for parameterization of complex shapes. Since 3D models are used, the shapes could also be parameterized using surface-based geometry definitions such as B-Spline or NURBS surfaces.

Acknowledgements

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Appendix A. Examples of intermediate designs

Figure 19 shows various designs which were included in the optimization design space. This illustrates the wide variety of designs that were analyzed to find the global optimum. Note that designs with 2,4 and 6 blades respectively represent separate optimization cases.

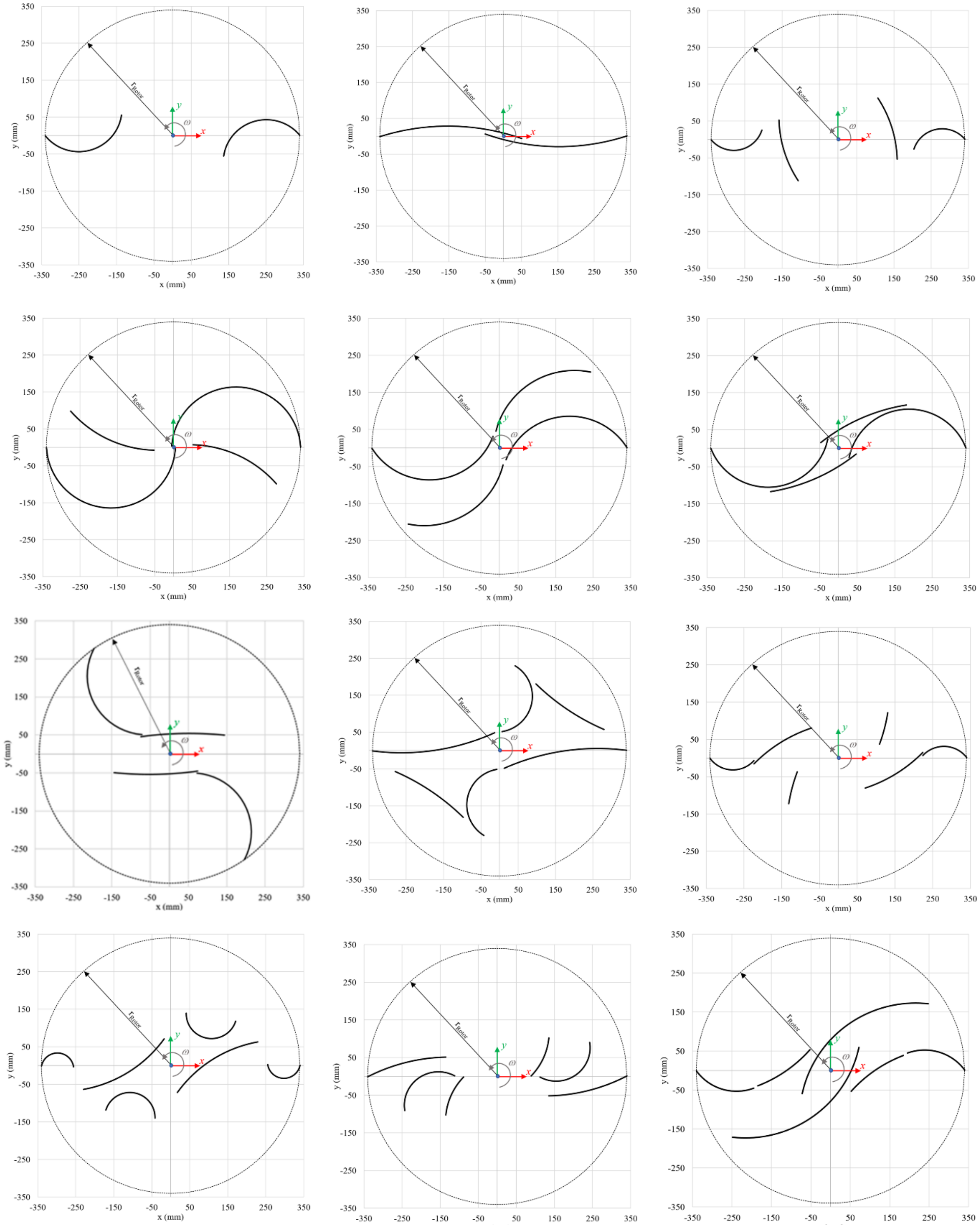

Figure 19. Examples of intermediate generated design.